Rajiv Gandhi University of Knowledge Technologies
(Department of MATHEMATICS)
MONTHLY TEST

Year/Sem:P2-SEM1-MID2
Subject: Mathematics
Subject Code: 20PMA2101

Date: 06-04-2022
Time: 45 Minutes
Max. Marks : 40 Marks

Answer any FOUR of the following questions. Each one carries 10 marks. (4X10M = 40M)

1. a) *If* $(1+x+x^2)^n = a_0 + a_1x + a_2x^2 + \ldots\ldots + a_{2n}x^{2n}$

   then show that $a_0 + a_3 + a_6 + a_9 + \ldots\ldots = 3^{n-1}$ (5M)

   b)find the middle term and the term independent of 'x' in the expansion of $\left(\frac{3}{2}x^2 - \frac{1}{3x}\right)^6$ (5M)

2. a) *If 'n' is a positive* integer *then prove that*

   $(i)\sum_{r=1}^{n} r.{}^nC_r = n.2^{n-1}$

   $(ii)\sum_{r=2}^{n} r(r-1){}^nC_r = n(n-1)2^{n-2}$

   use the above results to prove

   $2.{}^nC_0 + 5.{}^nC_1 + 8.{}^nC_2 + \ldots\ldots + (3n+2){}^nC_n = (3n+4)2^{n-1}$ (7M)

   b)find the numerically greatest term in the expansion of $(3-2x)^9$ when x=1 (3M)

3. a) Using binomial theorem, prove that "$81^n + 20n - 1$" is divisible by '100' for all positive integer values of 'n' (5M)

   b)If repetitions are not allowed, how many 3-digit numbers can be formed using the digits 0,1,2,3,……,9?

   How many of these numbers are divisible by 5?

   How many of these numbers are greater than 600? (5M)

4. a) How many different words can be formed by using all the letters of the word ALLAHABAD?

   In how many of them vowels occupy the even position?

   In how many of them both L do not come together? (5M)

   b)Find the sum of all 4-digit numbers that can be formed using the digits 1,3,5,7,9 .(repetitions are not allowed). (5M)

5. a)Find the number of ways of selecting 11 member cricket team from 7 bats men, 6 bowlers and 2 wicket keepers so that the team contain at-least 4 bowlers and one wicket keeper? (5M)

b)Find the number of ways of choosing 4 cards from a pack of 52 playing cards?
In how many of these selections all 4 cards are of same suit?
In how many of these selections 4 cards belongs to 4 different suits? (5M)

6. a)If $f(x) = x^3 - 3x^2 - 1$, for $x > 2$, then find the value of derivative of inverse function of f(x) at $x = -1 = f(3)$? (5M)

b) *If* $y = \dfrac{(1-2x)^{\frac{2}{3}}(1+3x)^{\frac{-3}{4}}}{(1-6x)^{\frac{5}{6}}(1+7x)^{\frac{-6}{7}}}$, *then find* $\dfrac{dy}{dx}$ using logarithmic *differentiation.* (5M)

7. a) *If* $y = \ln\left(\dfrac{\sqrt{1+e^x}-1}{\sqrt{1+e^x}+1}\right)$, *then show that* $\dfrac{dy}{dx} = \dfrac{1}{\sqrt{1+e^x}}$ (5M)

b) *show that* $\lim\limits_{x\to\infty}\left(1+\dfrac{3}{x}\right)^x = e^3$ (5M)

8. a) *show that* $\displaystyle\int_{1/2}^{1} \frac{6}{\sqrt{3+4t-4t^2}}\,dt = \frac{\pi}{2}$ (5M)

b) *Evaluate the* integral $\displaystyle\int_{5/4}^{2} \frac{dx}{1-x^2}$ *in terms of natural* logarithms. (5M)

Rajiv Gandhi University of Knowledge Technologies
(Department of Physics)
MONTHLY TEST

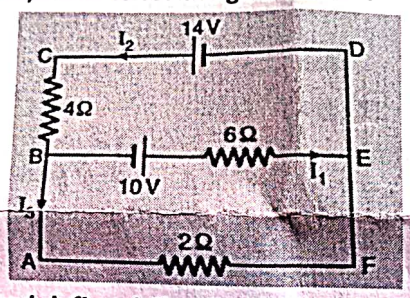
Year/Sem:P2-SEM1-MID2
Subject: PHYSICS
Subject Code:20PPY2101

Date: 07-04-2022
Time: 2 Hours
Max. Marks : 40 Marks

Answer any FOUR of the following questions. Each one carries 10 marks. (4X10M = 40M)

1. (a) State Kirchoff's laws in electricity and apply them to Wheatstone bridge to obtain balancing condition. (2+3+5)

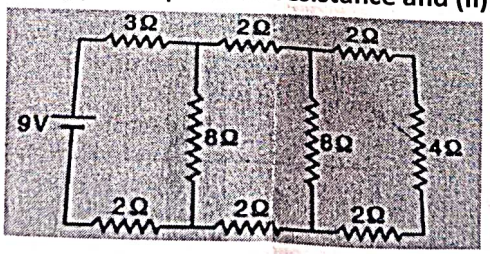
   (b) Find the currents I1,I2 and I3 for the given circuit.



2. (a)State Ohm's law and define drift velocity. (2+3+5)

   (b)Derive the expression for equivalent resistance when three resistors of resistances R1, R2 and R3 are connected in parallel.

   (c) For the given circuit, find (i) the equivalent resistance and (ii) power across battery.



3. (a) A straight conducting rod of length l is moved with a velocity v in a transverse magnetic field. Deduce the magnitude and direction of induced emf developed. If the ends of the rod are joined with a wire of resistance R, indicate the direction of induced current and its expression.

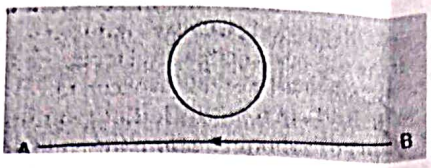
   (b) State Faraday's law and Lenz's law.

   (c)Find the emf induced in a coil of 200 turns and cross sectional area 0.2 m^2 when a magnetic field perpendicular to the plane of the coil changes from 0.1 T to 0.5 T at a uniform rate over a period of 0.05s. (5+2+3)

4. (a) Explain the phenomenon of Self induction and hence write the equation for self inductance (coefficient of self induction). (b) (2.5+2.5+3+2)

Explain the phenomenon of Mutual induction and hence write the equation for mutual inductance (coefficient of mutual induction). (c) Derive an expression for the self inductance of a solenoid of length l and consisting of N number of turns.

(d)The electric current in a wire in the direction from B to A is increasing with time. What is the direction of induced current in the circular loop kept above the wire as shown in figure.



5. (a) Derive an expression for growth of charge across cpaacitor in RC circuit and hence draw a graph how charge increases with time. (b) Define time costant and write its formula incase of RC circuit.

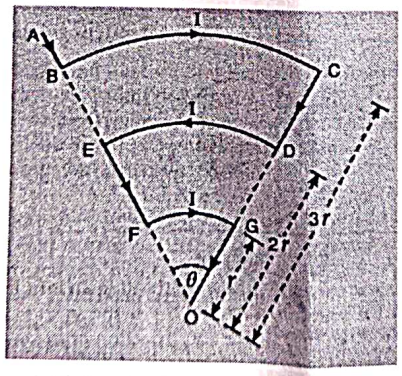
(c) A capacitor of 2 μF is connected to 20V supply through 1 megaohm resistance. How long will it take the capacitor to charge up 63.2% of its maximum charge. (5+2+3)

6. (a) State Ampere's law. Derive an expression for the magnetic field due to an infinitely long wire carrying current. (4+3+3)

(b) A current of 10A is flowing east to west in an infinitely long wire kept along east- west direction. Find the magnetic field in a horizontal plane at a distance (a) 20 cm North and (b) 10 cm south from the wire.

(c) An electron is moving northwards with a velocity of 10^7 m/s in a magnetic field of 3T directed eastwards. Calculate the magnitude and direction of magnetic force on electron.

7. (a) State and explain Biot-Savart's law and also mention its vector form. (b) A metallic wire is bent into the form as shown in figure and carries a current I. If point O is the common centre of all the three circular segments of the wires of radii r, 2r and 3r, find the magnitude and direction of net magnetic field at point O. (4+6)



8. (a) What is a solenoid. Write the expression for the magnetic field due to a solenoid at a point inside it carrying current. (2+5+3)

(b) Obtain an expression for torque acting on a rectangular current loop, when placed inclined at an angle θ with the direction of magnetic field B. (c) Calculate the torque on a 100 turn rectangular coil of length 40 cm and breadth 20 cm carrying current of 10A when palced in making an angle of 60^0 with a magnetic field of 5 T.

Rajiv Gandhi University of Knowledge Technologies

(Department of PUC)

MONTHLY TEST – 2

Year/Sem: PUC-II_SEM I –MID -2
Subject: Chemistry
Subject Code: 20PCY2101

Date: 08 -04-2022
Time: 2 hours
Max. Marks : 40 Marks

Answer any FOUR of the following questions (4 x 10 M = 40 Marks)

1. Write any two preparation methods of Ehyl chloride and its three chemical properties with equations. (4+6)M

2. a) Explain the Nucleophilic substitution reactions in alkyl halides 8M
   b) Write the structure of 1-chloro-2,2-dimethyl propane. 2M

3. Write any two preparation methods of chloro benzene and its three chemical properties with chemical equations. 10 M

4. Write any two preparations and chemical properties of alcohol with chemical equations. 10M

5. a) Explain the mechanism of dehydration of alcohols 4M
   b)How do you distinguish three types of alcohols 6M

6. Write about following reactions
   i) Reimer-Tiemann reaction(3M)
   ii)Kolbes reaction (3M)
   iii)Nitration of phenol with dil HNO_3 and conc HNO_3 (4M)

7. write any two preparations of Diborane and explain its structure (4+6)M

8. What is allotropy .Explain the allotropic forms of carbon diamond and Graphite. (2+4+4)M

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Rajiv Gandhi University of Knowledge Technologies
(Department of ENGLISH)
MONTHLY TEST

Year/Sem:P2-SEM1-MID2
Subject: ENGLISH
Subject Code: 20PEG2101

Date: 07-04-2022
Time: 2 Hours
Max. Marks : 40 Marks

Answer any FOUR of the following questions. Each one carries 10 marks. (4X10M = 40M)

1. How did J C Bose prove that plants experience pain like every other living being?
2. *Do* you prefer to gain knowledge through books or become wise through experience of Nature? Give reasons in support of your answer with reference to the poem *The Tables Turned*
3. Annotate the following lines:

   Build to-day, then, strong and sure,
   With a firm and ample base;
   And ascending and secure;
   Shall to-morrow find its place.

4. Give the character analysis of *The Tempest.*
5. Identify the error in the sentence and correct it.

   a. The streets of Nellore are narrower than Vijayawada.

   b. The three first stanzas of the poem are confusing.

   c. The Mona Lisa is an unique painting.

   d. To swim is a pleasant exercise.

   e. The stranger entered into the building.

   f. The man is mortal.

   g. Ram is intelligent than Shyam.

   h. Why you are laughing?

   i. Myself will come to your house.

   j. The driver of the car was a honest man.

6. Write a letter to your friend congratulating him/her on his/her success in class 11 board exam.

7. A. Match the following Foreign Phrases with their appropriate meaning:

1.	Bete -noire	(a)	The fashionable society.
2.	Amour - propre	(b)	A hateful person or thing.
3.	En -masse	(c)	Without preparation
4.	Beau - monde	(d)	Self-respect
5.	Ex-tempore	(e)	All together

B. Mark the stress for the following words.

1. Prefer
2. Survive
3. Technique
4. Among
5. Comparison

8. Find the odd word in the group which does not have a same sound (Homophone) to the sound of the word underlined:

1.	Call	Ce	Censor
2.	Unit	Upset	Umbrella
3.	Date	Drag	Rate
4.	One	Orange	Often
5.	Goat	Move	Goal
6.	Hid	Bid	Hide
7.	Look	Book	Mood
8.	Age	Stage	Begin
9.	Gum	Gem	Girl
10.	Care	Palm	Dare

Rajiv Gandhi University of Knowledge Technologies, Nuzvid Campus

(Department of Telugu)

PUC-2, SEM-1, MT -2 Telugu Question Paper

Subject: Telugu | **Time: 2 hours**

Code: 20P TE2101 | **EXAM DATE: 6-4-2022** | **Max marks: 40**

ఇచ్చిన వాటిలో నాలుగింటిని ఎంపిక చేసుకొని సందర్భ సహిత వ్యాఖ్యల్ని రాయాలి. 4 x 3 = 12 మార్కులు

Select four of the given ones and write contextual comments.. 4 x 3 = 12 marks

1. జాడలే దిచట సౌఖ్యం బెంత క్రీడించునో (Jāḍalē dicaṭa saukhyam benta krīḍinc̄unō)
2. అవని పాలించు భస్మ సింహాసనంబు (Avani pālinc̄u bhasma sinhāsanambu)
3. మృత్తికం గలిసెనో కద, కుమ్మరి వాని సారెపై(Mṛttikaṁ galisenō kada, kum'mari vāni sārepai)
4. లోకంబున వృద్ధి కాదగిన యేయే విద్యలల్లాడెనో(Lōkambuna vṛd'dhi kādagina yēyē vidyalallāḍenō)
5. యూరట గల్పించు నభేద భావమును ధర్మంబిందు గారాడెడిన్ (Yūraṭa galpimcu nabhēda bhāvamunu dharmambindu gārāḍeḍin)
6. కన్నీంటం బడి క్రాగి పోయినవి నిక్కంబిందు పాషాణముల్ (Kannīṭam baḍi krāgi pōyinavi nikkambindu pāṣāṇamul)

ఇచ్చిన 2 పద్యాల్లో 1 పద్యాన్ని పాదభంగం, అక్షర దోషాలు మొదలైనవి లేకుండా పూరించాలి 1 x 3 = 3 మార్కులు

Complete 1 of the 2 verses given without punctuation, typos etc. 1 x 3 = 3 marks

గమనిక: తెలుగు రాని విద్యార్థులు తప్పనిసరిగా పద్యాన్ని తెలుగులోనే రాయాలి

Note: Non-Telugu students must write the poem in Telugu

7. ఎన్నో యేండ్లు గతించి.. (Ennō yē0ḍlu gatinc̄i)
8. ఇట నస్పృశ్యత సంచరించుటకు.. (Iṭa nasprṣyata sanc̄arinc̄uṭaku)

ఇచ్చిన ప్రశ్నల్లో ఏవైనా ఐదు ప్రశ్నలకు వ్యాసరూప సమాధానమివ్వండి 5 x 5 = 25 మార్కులు

Answer any five of the given questions in essay format 5 x 5 = 25 marks

9. వినాయక చవితి గొప్పతనాన్ని వివరించండి (Explain the greatness of Vinayaka Chavati)
10. సంక్రాంతి పండగ విశిష్టతను పేర్కొనండి (Specify the uniqueness of Sankranthi festival)
11. పండుగల ప్రాముఖ్యతను ఎలా అర్థం చేసుకోవాలో తెలపండి(Let us know how to understand the importance of festivals)
12. కలవారి కోడలూ కలికి కామాక్షి పాఠ్య భాగ నేపథ్యాన్ని వివరించండి(Explain the background of the Kalavāri kōḍalū kaliki kāmākṣi lesson)
13. కలవారి కోడలూ కలికి కామాక్షి లో అన్నా చెల్లెళ్ల అనుబంధం ఎలాంటిది(explain the brother sister relation in Kalavāri kōḍalū kaliki kāmākṣi lession
14. క్రిస్ మస్ పండుగను విశ్లేషించండి (Analyze Christmas)
15. కలవారి కోడలూ కలికి కామాక్షి లో గ్రామీణ సంస్కృతిని సమీక్షించండి(Review the rural culture of Kalavāri kōḍalū kaliki kāmākṣi lesion)

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Rajiv Gandhi University of Knowledge Technologies
(Department of INFORMATION TECHNOLOGY - PUC)

MID 2 Examination – 2020 Batch

Year/Sem: P2-SEM1-MID-2
Subject: W D with HTML, CSS and Intro to Programming
Subject Code: 20PIT2101

Date: 08-04-2022
Time: 2 hours
Max. Marks: 40 Marks

Answer any FOUR from the following questions, each carries 10 Marks.

1) List some advantages and disadvantages of CSS. (5M+5M)

2) Explain about text properties used in CSS.

3) Write an HTML code with at least 4 tags. Apply External styling for those 4 tags.
(Have to use 4 different CSS properties for 4 HTML different tags)

4) Write about JavaScript History and Applications of JavaScript (4M+6M)

5) Write a JavaScript program to find the relation between to numbers. (Numbers should be taken from the user. You can use any method to take input or to print output).

6) Explain about symbols used in flowchart.

7) Write an algorithm to find a number is even or odd.

8) Draw a flowchart to find the number of factors for a given number.